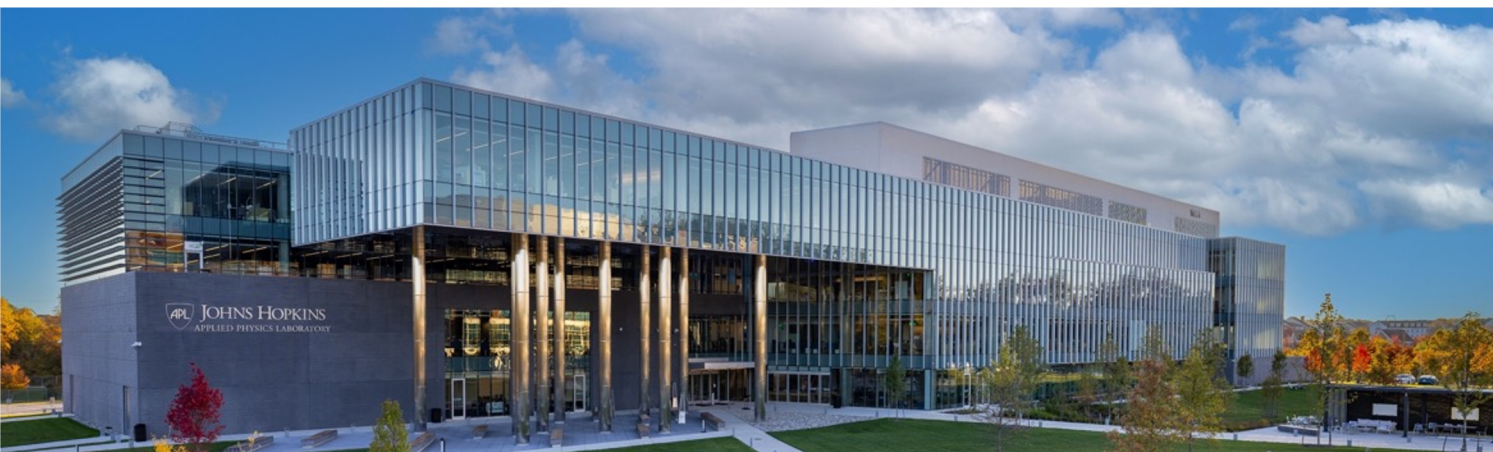
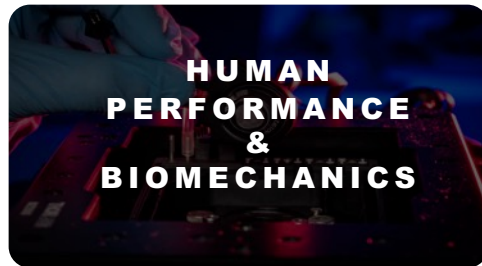
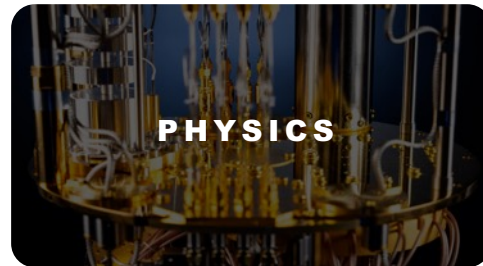


AI for Science at JHU/APL: Access, Execution and Verification at the Jagged Frontier

Mike Wolmetz, PhD
Program Manager
Frontier Intelligent Systems

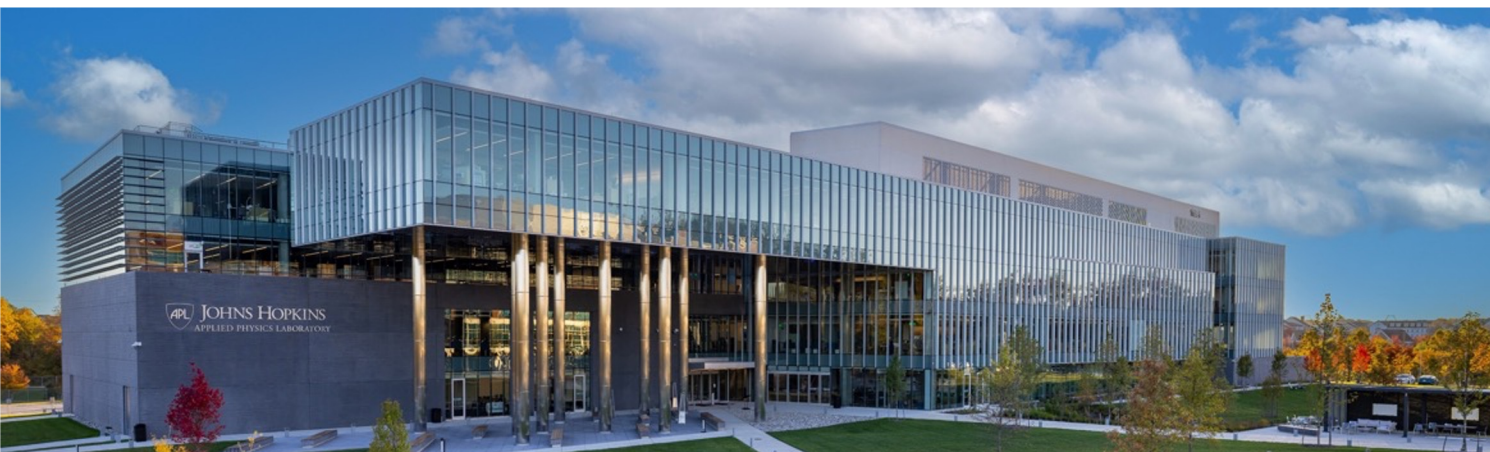
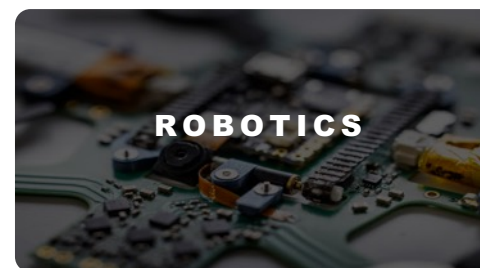
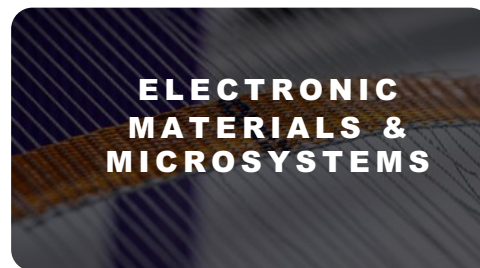
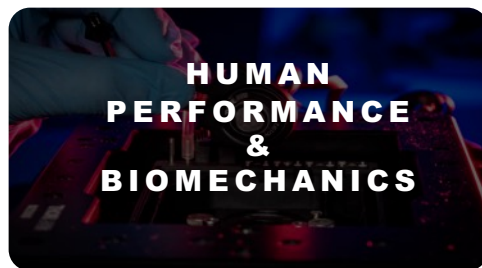
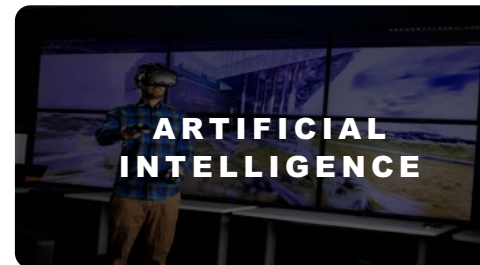
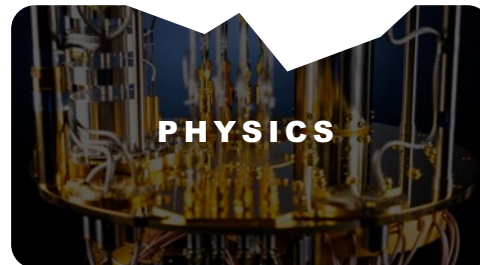
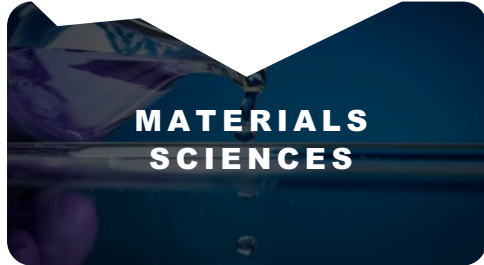
michael.wolmetz@jhuapl.edu
michael.wolmetz@coe.ic.gov (JWICS)

Science is Changing Rapidly but Jaggedly



Research and Exploratory Development Department

Science is Changing Rapidly but Jaggedly



Research and Exploratory Development Department



Intelligent Systems Center 2.0 (2023)

Why the AI for Science Frontier is Jagged



1 Access

AI Enabler

Interact with relevant literature, data, prior results, models, tools, and context



2 Execution

Run simulations or safely invoke laboratory actions through a controlled interface



3 Verification

Produce results with evidence that can be inspected, interpreted, and verified at scale

AI Limiter

*Most laboratory science involves significant **tacit knowledge** that may not be accessible*

*Some domains are more amenable to **simulation** and **lab automation** than others*

***Efficient verification** via physical measurement, simulation, or expert assessment is a bottleneck*



1 Access

TASK: Prepare a *Bacillus thuringiensis* spore preparation and quantify its purity. Describe what the results will look like.

NON-EXPERT

4 / 18

“NON-VIABLE”

Title: No Research B. thuringiensis
Author: Subject 2
Date: Sept. 22nd, 2023
Time: 10 mins
Task: Prepare a *Bacillus thuringiensis* spore preparation and quantify its purity. (Describe what the results will look like.)

Yellow highlight from information gleaned from previous beta-testing

Materials:

- PPE: gloves, eye protection, lab coat
- Culture of *Bacillus thuringiensis*
- Spore stain

- Sporulation medium
- Incubator
- Microscope
- Slides
- Slide Rack
- Inoculation Loops
- Culturing tubes
- Spore stain

Procedure:

1. Inoculate growth medium with sample of *B. thuringiensis*
2. Incubate a STEMP for \$TIME, look for \$SIGNS_OF_GROWTH
3. Inoculate sporulation medium
4. Incubate at STEMP for \$TIME
5. Stain sporulated culture with spore stain, following SPROCEDURES
6. Observe under microscope, spores should be stained green
7. To quantify purity, follow SPROCEDURES

Verification:
1. ???

NON-EXPERT + LLM

13.5 / 18

“Extremely short protocol that would require an extreme amount of tacit knowledge to complete successfully.”

Materials:

- *Bacillus thuringiensis* (Sigma-Aldrich, 100-1000 CFU)
- Nutrient-rich media (Sigma-Aldrich, 100-1000 CFU)
- Sporulation media (Sigma-Aldrich, 100-1000 CFU)
- Inoculation Loops (Sigma-Aldrich)
- Culturing tubes (Sigma-Aldrich)
- Microscope slides
- Personal protective equipment - Lab coat, gloves, and goggles

Sporulation Induction

Materials:

- *Bacillus thuringiensis* (Sigma-Aldrich, 100-1000 CFU)
- Nutrient-rich media (Sigma-Aldrich, 100-1000 CFU)
- Sporulation media (Sigma-Aldrich, 100-1000 CFU)
- Inoculation Loops (Sigma-Aldrich)
- Culturing tubes (Sigma-Aldrich)
- Microscope slides
- Personal protective equipment - Lab coat, gloves, and goggles

Procedure

1. Inoculate *Bacillus thuringiensis* into the nutrient-rich media.
2. Incubate at 37°C for *Bacillus* for 24 hours
3. Transfer the culture to sporulation medium.
4. Incubate the cultures 30°C for 5-7 days
5. Prepare a smear by spreading a small drop of the clean, grease-free microscopic slide.
6. Examine the culture under a phase-contrast microscope.

Observations:

Bacillus thuringiensis spores under a phase contrast microscope appear as bright, rod-shaped cells. Spores appear brighter due to their higher refractive index. The spores can be central, subterminal, or terminal, and they are typically oval or cylindrical.

The vegetative cells can vary in length but are typically between 1-2 micrometers in diameter. They are rod-shaped and have a distinct structure.

Verification:

Successful sporulation should result in the majority of cells associated with spore formation—spore purity assessment.

Procedure

1. Prepare a bacterial smear by spreading a small drop of the sporulated *Bacillus* culture across a clean, grease-free microscopic slide.
2. Allow the smear to air-dry completely. Then, pass the slide through a Bunsen burner flame several times (smear side up) to heat fix the smear onto the slide.
3. Prepare a boiling water bath.
4. Cover the slide with a piece of paper towel and place on staining rack over the water bath.
5. Flood the paper towel with primary stain
6. Steam slide for 5 to 7 minutes
7. Remove paper towel and allow to cool
8. Rinse the slide with water for 30 seconds or until water runs clear.
9. flood the smear with the counterstain, safranin, and wait for 1-2 minutes. Rinse again thoroughly.
10. Gently blot dry and view the slide under the microscope. Start from a lower magnification to locate your bacterial cells then switch to the oil immersion objective (usually 100x) for detailed observation and counting.

Observations:

After staining, the spores (which retain the malachite green stain) should appear green and the vegetative cells (which take the safranin stain) should appear red. You should count the total number of cells (both red and green) and the number of green spores.

Verification:

Divide the number of spores (green cells) by the total number of cells (both green and red) multiplied by 100 to get the percentage of spores. This gives an estimate of the purity of sporulation. A high percentage (generally, above 80-90%) indicates a high purity of sporulation.

Why the AI for Science Frontier is Jagged



1 Access

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2 Execution

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Starting to Close the Loop in Materials

SOTA in 2023

Superconductor discovery



1 Access

npj | computational materials

www.nature.com/npjcompumats

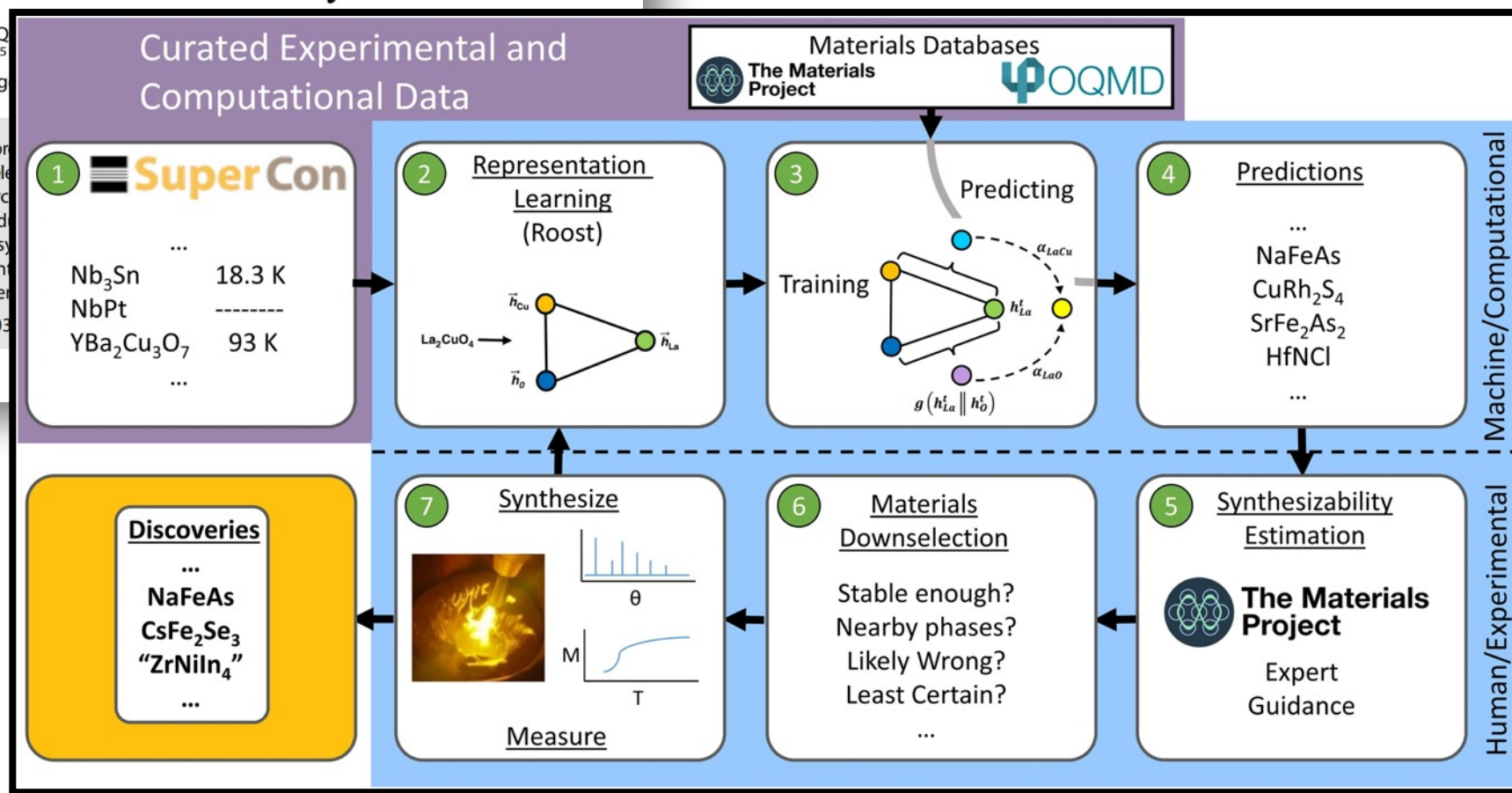
ARTICLE OPEN

Closed-loop superconducting materials discovery

Elizabeth A. Pogue^{1,6}, Alexander New^{1,6}, Kyle McElroy¹, Nam Q. Janna Domenico¹, Elizabeth Hedrick³, Tyrel M. McQueen^{3,4,5}, Andrew Lennon¹, Christine Chung¹, Timothy Montalbano¹, Greg

Discovery of novel materials is slow but necessary for societal progress. A machine learning (ML)-driven approach to rapidly explore a large materials search space, accelerated by experimentally validating the results of the ML-generated superconductor model to refine, we demonstrate that success rates for superconducting materials discovery are significantly improved. In this work, we report discovery of a superconductor in the Zr-In-Ni system, and identification of two additional phase diagrams of interest. This work highlights the critical role experimental feedback provides in ML-driven discovery.

npj Computational Materials (2023)9:181; <https://doi.org/10.1038/s41524-023-01181-1>



Starting to Close the Loop in Materials

SOTA in 2023

Superconductor discovery



1 Access

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materials

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Check for updates

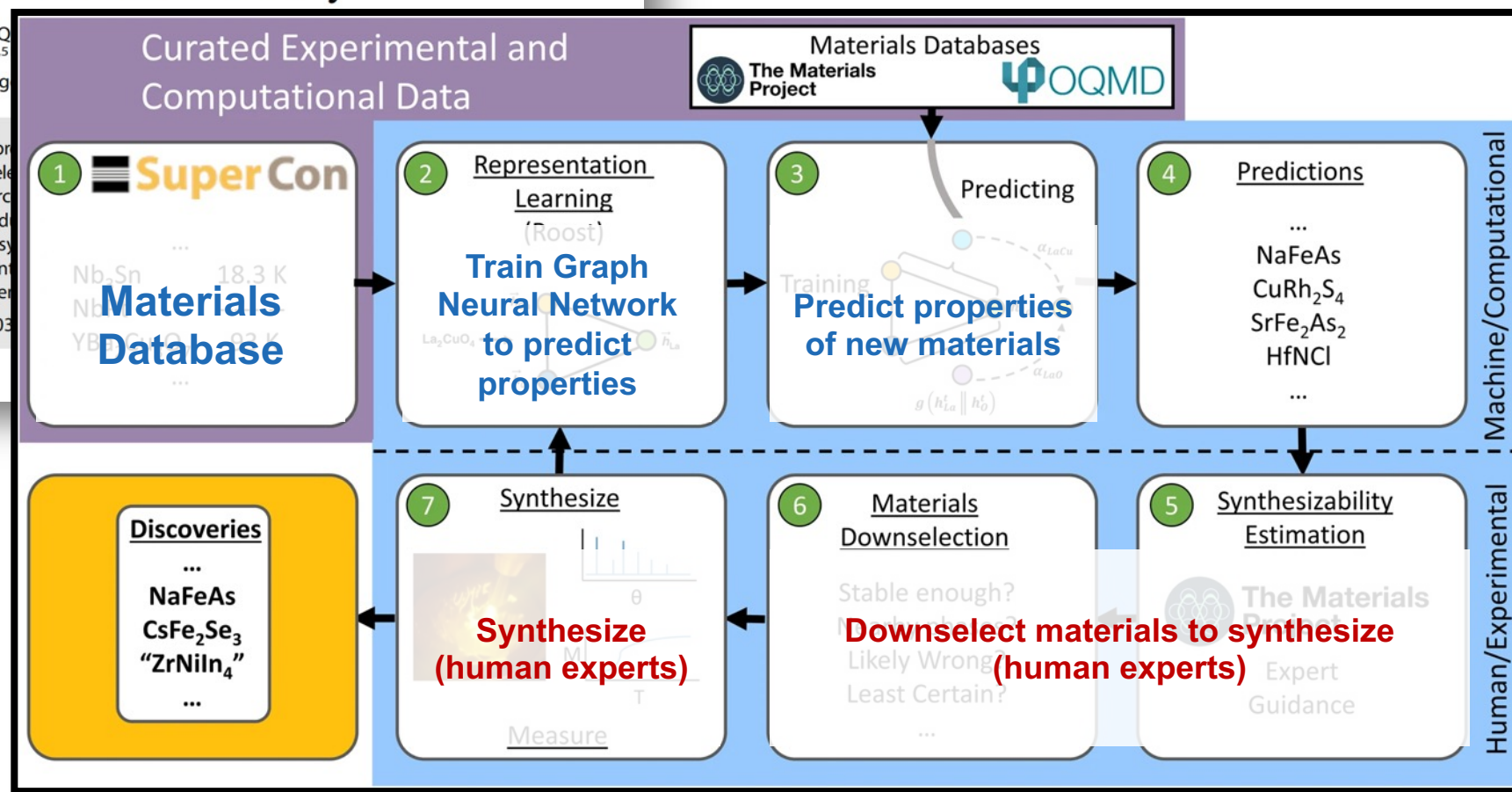
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npj Computational Materials (2023)9:181; <https://doi.org/10.1038/s41524-023-01181-1>



Autonomous Closed-Loop Synthesis and Characterization

SOTA Today
Battery Electrolyte Discovery



1 Access



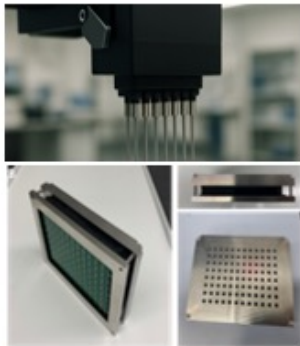
2 Execution

“Generate a hypothesis and test plan to improve low temperature performance of lithium ion batteries through new electrolyte design”

1. Candidate Selection



2. Sample Preparation



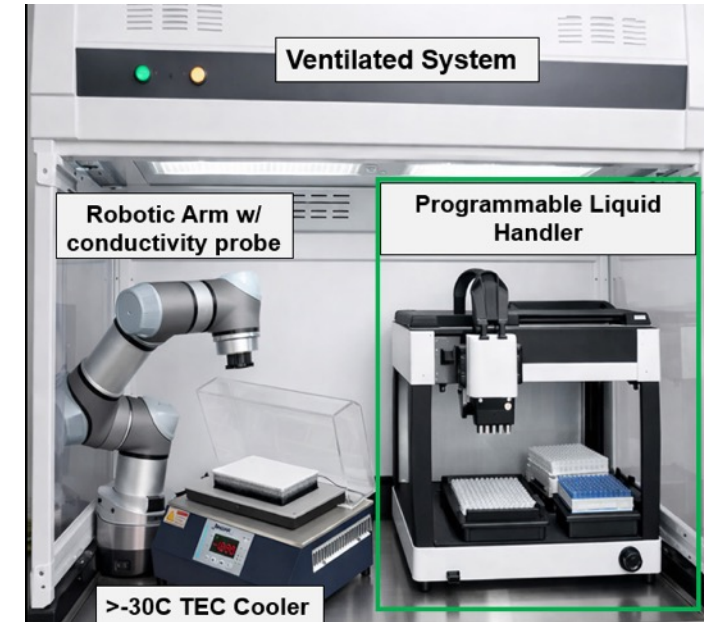
3. In-Situ Characterization



4. Data Management & Assessment



5. Decision-Making

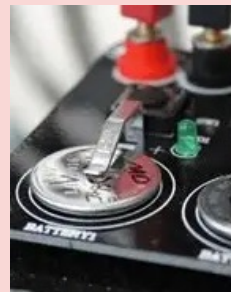


Outcomes

- Designed and autonomously screened 485 battery electrolyte formulations
- System reduces researcher effort from ~44 hours of manual experimentation to ~1.5 hours experiment design and system setup.
- Top candidate delivers over twice the capacity of leading benchmarks at -40°C

AI Limiter

- Final testing done manually by constructing cells with each electrolyte
- Requires dexterity and judgment beyond today's robotic autonomy



Autonomous Closed-Loop Synthesis and Characterization

SOTA Today
Microcapsule Discovery

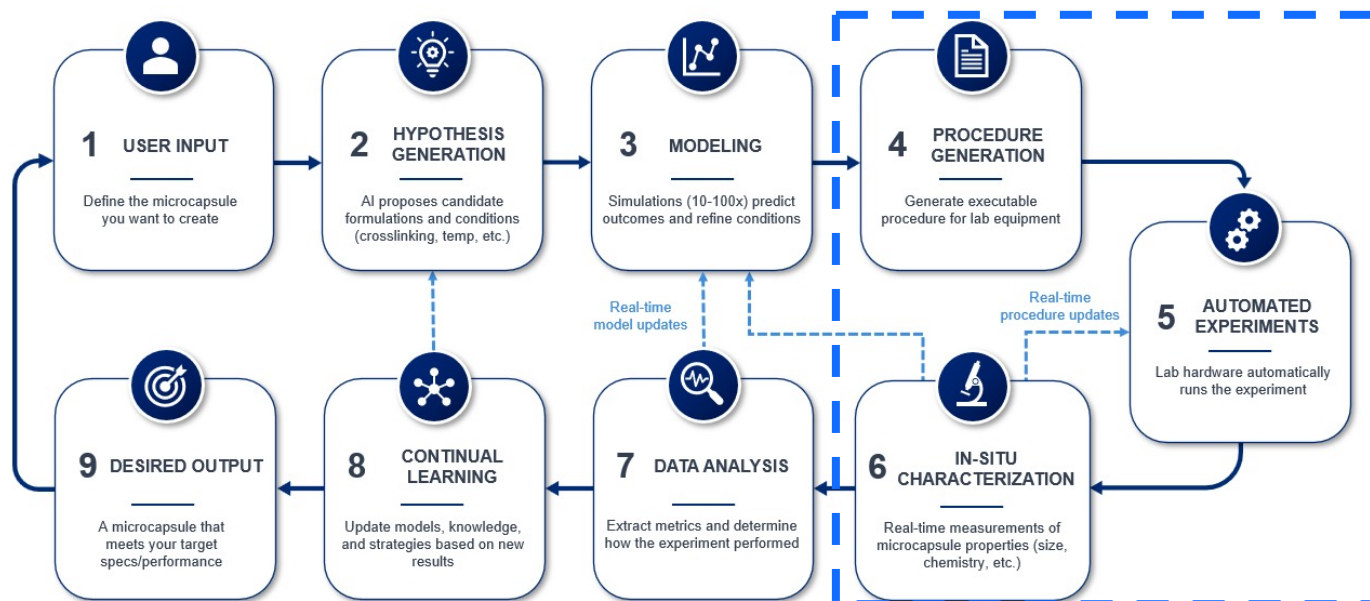


1 Access

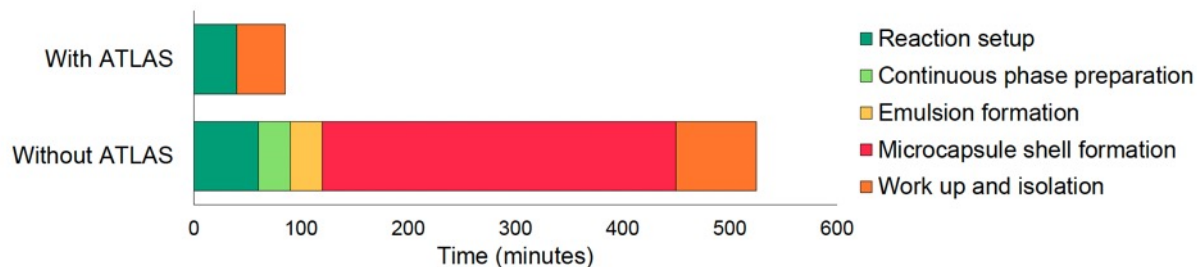


2 Execution

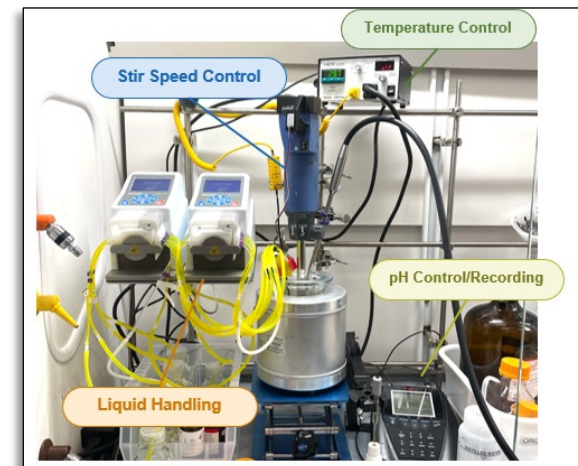
“Make a microcapsule that encapsulates mineral oil and is robust to spraying”



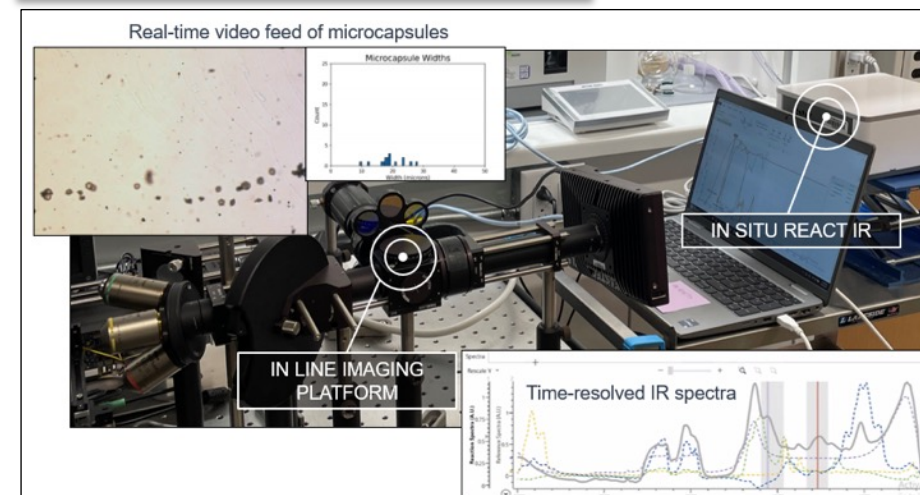
Timeline of Human Effort: With vs. Without ATLAS



Synthesis



In-situ Characterization



Why the AI for Science Frontier is Jagged



1 Access



2 Execution



3 Verification

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***Efficient verification** via physical measurement, simulation, or expert assessment is a bottleneck*

Human and AI SMEs



1 Access 3 Verification

Sample materials science claim: *"An aluminum alloy with thermally stable nanometer-scale dispersoids can maintain microstructures with grain sizes of 10 microns or less during superplastic forming, enabling elongation beyond 300% without premature loss of ductility."*

SFBench: The SciFy Scientific Feasibility Benchmark

Cash Costello, James Mayfield, Elsbeth Turcan, Christine Piatko, Christina K. Pikas, Justin Rokisky, Sam Scheck, Chris Ribaud, Ritwik Bose, Alex Memory
Johns Hopkins Applied Physics Laboratory
11100 Johns Hopkins Rd.
Laurel, MD 21045

Abstract

We present SFBench, a benchmark dataset for evaluating systems that assess the feasibility of scientific claims. SFBench includes 197 claims in materials science, each annotated with a ground-truth feasibility score on a five-point scale along with an explanation of that assessment. The collection differs from previous collections in several important ways: 1) it defines a complex task that requires reasoning over claims of varying scientific feasibility; 2) its claims are not extracted from existing scientific publications but are created de novo, greatly reducing the chances that LLMs have trained on them; 3) claims and ground truth are established by subject matter experts, not by artificial intelligence; and 4) unlike many benchmarks that ask about question/answer pairs, provide multiple choice answers, or ask questions requiring short, fixed answers, SFBench explanations are completely open-ended. We describe the benchmark design, data creation process, and evaluation metrics, and we report baseline results using recent GPT models.

1 Introduction

This paper describes SFBench, a new benchmark for systems that assess feasibility of scientific claims. A scientific claim such as "a perovskite solar cell of ABX₃ composition was formed with

SFBench comprises a set of scientific claims in the domain of materials science together with ground truth feasibility assessments of those claims that can be used to compare the effectiveness of systems that are evaluated on these claims.

In practice, feasibility assessment is typically performed by human experts, who synthesize evidence from prior literature, theoretical constraints, and available technologies to judge whether a claim can be realized. This process is time-consuming, requires deep domain expertise, and does not scale well as the volume of scientific claims increases. These challenges motivate the development of automated systems that can assist with, or partially automate, scientific feasibility assessment.

As scientific claims proliferate in press releases, social media, and scientific publications, the ability to assess their feasibility is increasingly important. Accurate feasibility assessment is important for research prioritization, investment evaluation, technology forecasting, and policy planning. At the same time, infeasible claims can arise for many reasons. A claimant might mistakenly believe a claim to be true. For example, a journalist may misunderstand the results of a scientific paper, or a scientist may overlook a confounding factor or calculation error. Claims can also originate from large language models, which may hallucinate plausible-

April 14, 2026 | 5 min read | Add Us On Google

DARPA's AI is built to call BS on wild weapons claims

SCIENTIFIC AMERICAN

The SciFy program tests whether adversaries' most outlandish scientific claims add up or fall apart

BY SARAH SCOLES | EDITED BY ERIC SULLIVAN



Richard Mila

[SFBench: The SciFy Scientific Feasibility Benchmark](#)

[DARPA built an AI to fact-check enemy weapons claims | SciAm](#)

Automating Replication



1 Access



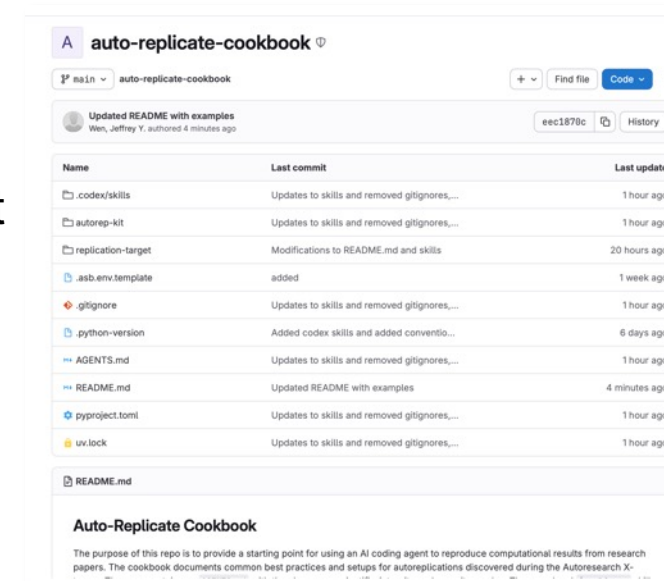
2 Execution



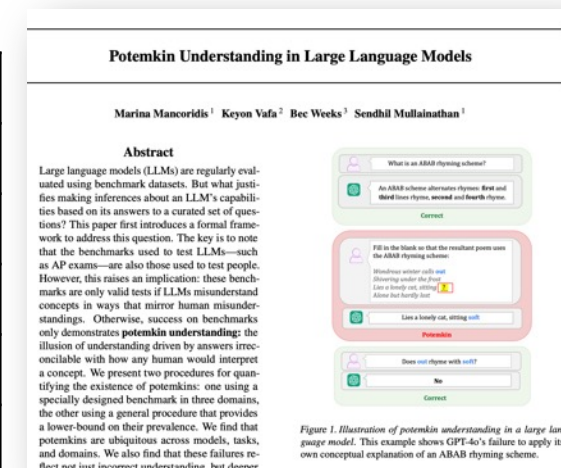
3 Verification

AutoReplication Cookbook v0.1

1. Ingest material: load the paper, code, and data
2. Select Claim: pick the specific result to test
3. Plan: map out how to test it
4. Implement: write or adapt the code
5. Test/validate: run it and compare to the paper
6. Aggregate & Report: summarize matches, gaps, and assumptions



Category	Papers
AI/ML, agents, and robotics	15
Statistics/economics/social science	5
Physical science/materials	3
Numerical/scientific computing	1
Classical computer vision	1



ISCONNECT EVENT

SEPTEMBER 8

AutoReplicate Hack-a-Thon

Use AI agents to replicate real scientific results - in one day.

Paper

select a publication

Agent

launch an AI workflow

Replicate

reproduce a result

Extend

push it further

What you will do

Pick a paper.
Identify a result that can be replicated.

Use an AI agent.
Build, run, debug, and document the effort.

Share the story.
What worked? What failed? What did you learn?

Why join?

Hands-on learning.
Try AI-agent workflows in a practical setting.

Support available.
Office hours throughout the day.

Time + tokens covered.
All participants get a budget to cover both.

Schedule

9:00 AM Launch + rules + publication selection

10:00 AM Getting Started with AI Agents at APL Workshop: sandbox + Portkey

All day Office hours and support

Afternoon Report-out: attempts, lessons, and next steps

Bring your curiosity. No perfect replication plan required.

Hosted by the AutoResearch X-Team

Access, execution, and verification for every domain



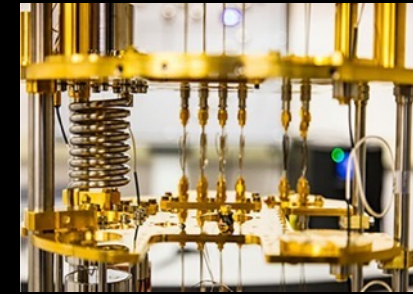
Operational Marine Bio



Additive Manufacturing Center



Advanced Optics and Photonics Laboratory



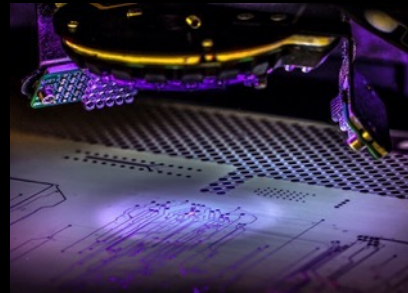
Low Temperature Quantum Device Laboratory



Vertically Accelerated Load Transfer System



Plant Growth Laboratory



Advanced Electrical Fabrication Laboratory



Genome Sequencing Laboratory



Materials Characterization Facility



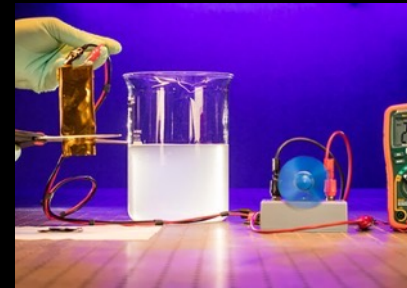
Intelligent Systems Center



Materials for Extreme Environments Laboratory



Mechanical Fabrication Laboratory



Energy Storage Materials Laboratory



Semiconductor Thermoelectrics Applications and Research



JOHNS HOPKINS
APPLIED PHYSICS LABORATORY